A template/ approach to Reproducible Generative Research

Research infrastructure that documents itself — a case study for meta-science and science-integrity teams

The problem

- Research groups publish claims about their own tools and pipelines that no one outside the team can re-check.
- Manuscripts describing infrastructure go stale the moment the infrastructure changes underneath them.
- Reproducibility efforts usually stop at the data and code layer — the documentation layer is still hand-maintained and drifts silently.

What template_template is

• A manuscript that is generated FROM the repository it describes, not written ABOUT it from memory.

• Every metric in the paper — module counts, test counts, pipeline stages — is computed live by template_template's own introspection code.

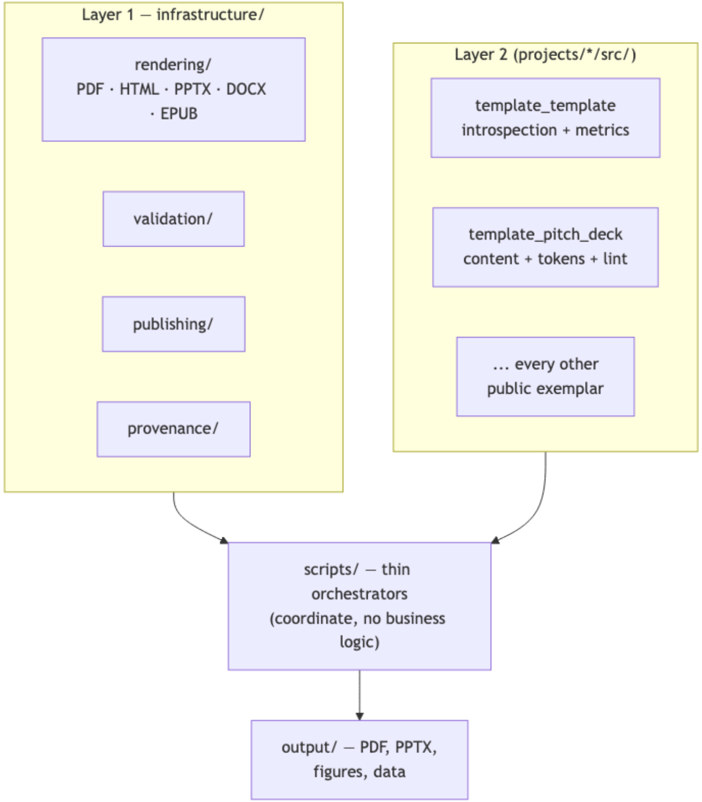
• One of 24 public exemplar templates in the same monorepo, each independently tested and coverage-gated.

Source: infrastructure/project/public_scope.py

How it works

- src/template_template/introspection.py scans infrastructure/ modules, the pipeline DAG, and the public project roster.
- metrics.py turns that scan into a dictionary of ${variable} values; inject_metrics.py substitutes them into the manuscript text.
- Re-running the pipeline regenerates the paper from current repository state — the citation and the artifact cannot drift apart.

Two-layer architecture, at a glance



Source: CLAUDE.md

Proof, not adjectives

143 tests

99.14% coverage on template_template's own source — the same gate every exemplar in the repo must pass.

Source: docs/_generated/COUNTS.md

Already public

- Concept DOI 10.5281/zenodo.20419007, 9 durable publication records already on file: Zenodo, GitHub, PyPI (sandbox), IPFS, Software Heritage, GitHub Pages, Netlify, Hugging Face, OSF.
- Licensed Apache-2.0 — the introspection code, not only the prose, is available for a science-integrity team to audit directly.

Source: projects/templates/template_template/manuscript/config.yaml

Why this matters for science integrity

- A funder or reviewer can clone the repository and regenerate the exact figures and counts cited in the manuscript, on their own machine.
- The same no-mocks, coverage-gated testing discipline that verifies the code also verifies the claims made about the code.
- The pattern generalizes across the whole projects/templates/ folder — all 24 exemplars, not just one cherry-picked example — follow the identical thin-orchestrator, two-layer architecture.

Source: docs/_generated/active_projects.md

Cite this deck

- This pitch deck (template_template_pitch_short.pdf) is its own exemplar in the same monorepo — it has its own source, tests, and citable identity, separate from the DOI of the project it pitches.
- This deck's own DOI: 10.5281/zenodo.21281509.
- Every slide also carries a QR code to its own standalone, citable page — see the deep-linking mechanism described later in the medium and long versions of this deck.

Source: projects/templates/template_pitch_deck/manuscript/config.yaml

The ask

- Adopt this pattern for your own methods papers: bind every reported number to the code that produced it, not to a hand-typed table.
- Pilot it on one existing manuscript in your group before committing to a full rewrite.
- Happy to pair with your team through a first regeneration cycle, if useful — no prior engagements to point to yet, this pattern is newly forkable.
- Open to funding conversations and scoping a consulting engagement for a deeper or custom integration — no prior engagements exist yet; this is an invitation to talk, not an established offering.

Questions

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